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RFC 9562
Universally Unique IDentifiers (UUIDs)

Abstract

This specification defines UUIDs (Universally Unique IDentifiers) -- also known as GUIDs (Globally Unique IDentifiers) -- and a Uniform Resource Name namespace for UUIDs. A UUID is 128 bits long and is intended to guarantee uniqueness across space and time. UUIDs were originally used in the Apollo Network Computing System (NCS), later in the Open Software Foundation's (OSF's) Distributed Computing Environment (DCE), and then in Microsoft Windows platforms.

This specification is derived from the OSF DCE specification with the kind permission of the OSF (now known as "The Open Group"). Information from earlier versions of the OSF DCE specification have been incorporated into this document. This document obsoletes RFC 4122.

Status of This Memo

This is an Internet Standards Track document.

This document is a product of the Internet Engineering Task Force (IETF). It represents the consensus of the IETF community. It has received public review and has been approved for publication by the Internet Engineering Steering Group (IESG). Further information on Internet Standards is available in Section 2 of RFC 7841.

Information about the current status of this document, any errata, and how to provide feedback on it may be obtained at https://www.rfc-editor.org/info/rfc9562.

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MSB0	MSB1	MSB2	MSB3	Version	Description
1	0	1	0	10	Reserved for future definition.
1	0	1	1	11	Reserved for future definition.
1	1	0	0	12	Reserved for future definition.
1	1	0	1	13	Reserved for future definition.
1	1	1	0	14	Reserved for future definition.
1	1	1	1	15	Reserved for future definition.

Table 2: UUID Variant 10xx Versions Defined by This Specification

An example version/variant layout for UUIDv4 follows the table where "M" represents the version placement for the hexadecimal representation of 0x4 (0b0100) and the "N" represents the variant placement for one of the four possible hexadecimal representation of variant 10xx: 0x8 (0b1000), 0x9 (0b1001), 0xA (0b1010), 0xB (0b1011).

```
00000000-0000-4000-8000-000000000000
00000000-0000-4000-9000-000000000000
00000000-0000-4000-A000-000000000000
00000000-0000-4000-B000-000000000000
xxxxxxxx-xxxx-Mxxx-Nxxx-xxxxxxxxxxxx
```

Figure 5: UUIDv4 Variant Examples

It should be noted that the other remaining UUID variants found in Table 1 leverage different sub-typing or versioning mechanisms. The recording and definition of the remaining UUID variant and sub-typing combinations are outside of the scope of this document.

5. UUID Layouts

To minimize confusion about bit assignments within octets and among differing versions, the UUID record definition is provided as a grouping of fields within a bit layout consisting of four octets per row. The fields are presented with the most significant one first.

5.1. UUID Version 1

UUIDv1 is a time-based UUID featuring a 60-bit timestamp represented by Coordinated Universal Time (UTC) as a count of 100-nanosecond intervals since 00:00:00.00, 15 October 1582 (the date of Gregorian reform to the Christian calendar).

UUIDv1 also features a clock sequence field that is used to help avoid duplicates that could arise when the clock is set backwards in time or if the Node ID changes.